ENG ECON

PRESENTED BY:

ANGELA FRANCESKA L. VILLAFUERTE
BSCPE 2-2
S 5:30PM - 8:30PM

PRESENTED TO:

PROF. JESSE WENCESLAO R. NIDEA

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CAPITALIZED COST

DEFINITION

- CAPITALIZED COST (CC) IS THE PRESENT WORTH OF A PROJECT THAT HAS A VERY LONG LIFE OR WHEN THE PLANNING HORIZON IS CONSIDERED VERY LONG OR INFI NITE.
- A CAPITALIZED COST IS A COST THAT IS INCURRED FROM THE PURCHASE OF A FIXED ASSET THAT IS EXPECTED TO DIRECTLY PRODUCE AN ECONOMIC BENEFIT BEYOND ONE YEAR OR A COMPANY'S NORMAL OPERATING CYCLE.

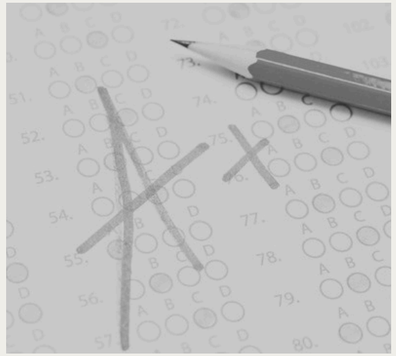
FORMULA

$$CC = FC + \frac{A}{i} + \frac{RC - SV}{(1+i)^L - 1}$$

WHERE:

CC	Capitalized Cost	L	Economic or Useful Life
A	Operating and Maintenance Cost	FC	First Cost
RC	Reconstruction Cost (if not stated, RC=FC)	i	Interest Rate
		SV	Salvage Value

EXERCISES



EASY

A BRIDGE COSTS $10M TO BUILD. ANNUAL MAINTENANCE IS $50,000. IF INTEREST IS 6%, WHAT IS THE CAPITALIZED COST?

MEDIUM

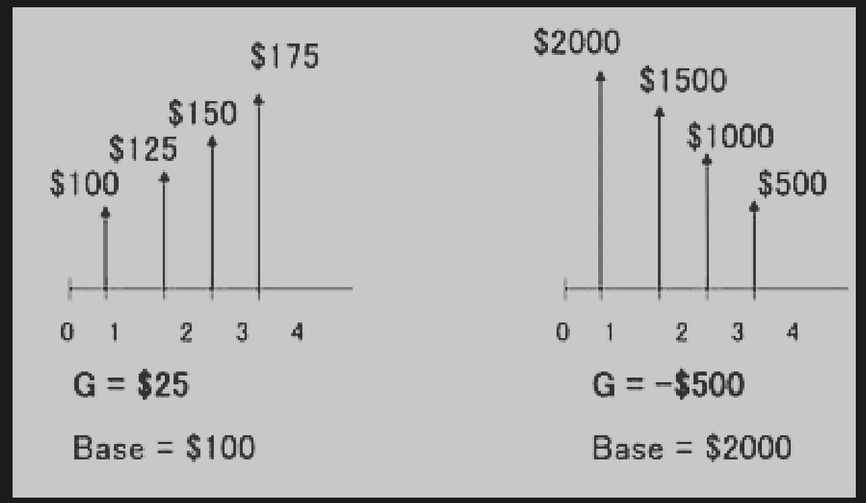
A MACHINE COSTS $500K, LASTS 10 YEARS, AND HAS A $50K SALVAGE VALUE. INTEREST IS 8%. COMPUTE THE CC.

CLINCHER

COMPARE TWO PAVEMENT TYPES FOR A HIGHWAY:

- CONCRETE: $12M INITIAL COST, LASTS 20 YEARS, $20K ANNUAL MAINTENANCE.
- ASPHALT: $5M INITIAL COST, LASTS 8 YEARS, $80K ANNUAL MAINTENANCE.
- WHICH IS MORE ECONOMICAL AT 10% INTEREST?

GRADIENT



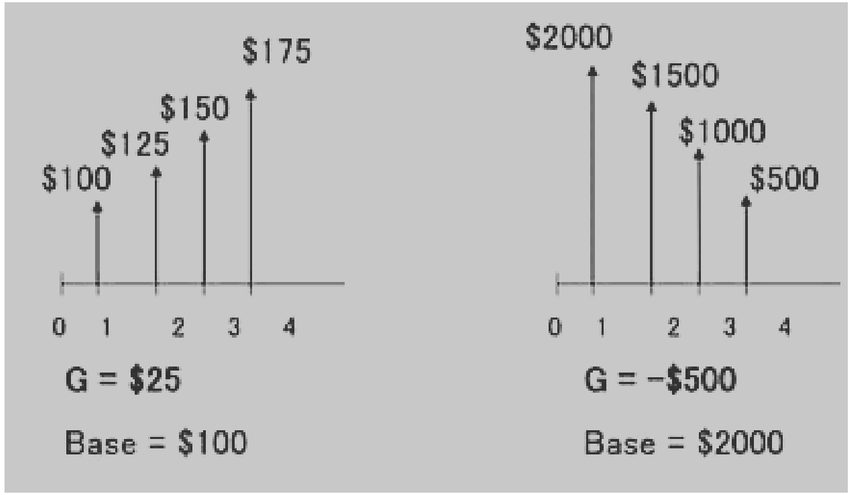
ARITHMETIC GRADIENT

- AN ARITHMETIC GRADIENT IS A CASH FLOW THAT EITHER INCREASES OR DECREASES BY A CONSTANT AMOUNT.4
- THE CASH FLOW CHANGES BY THE SAME AMOUNT EACH PERIOD.
- THE AMOUNT OF DECREASE OR THE INCREASE IS THE GRADIENT OR G.
- THEREFORE, IT IS COMPOSED OF THE BASE AMOUNT AND GRADIENT PART.

GEOMETRIC GRADIENT

- CASH FLOW SERIES THAT INCREASE OR DECREASE FROM PERIOD TO PERIOD BY CONSTANT PERCENTAGE.
- G IS THE CONSTANT RATE OF CHANGE IN DECIMAL FORM, BY WHICH AMOUNTS CHANGE
- THE SERIES STARTS AT THE END OF YEAR 1, WITH AMOUNT A1, WHICH IS NOT CONSIDERED AS A BASE AMOUNT

GRADIENT



ARITHMETIC GRADIENT FORMULA

$$CF_n = base\ amount + (n-1)G$$

$$A = G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$$

$$F = G\left[\frac{1}{i}\left(\frac{(1+i)^n - 1}{i} - n\right)\right]$$

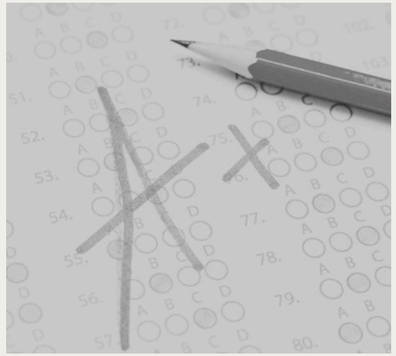
$$P_T = P_A \pm P_G$$

GEOMETRIC GRADIENT FORMULA

$$P_g = A_1\left[\frac{\left[1 - \frac{(1+g)^n}{(1+i)^n}\right]}{(i-g)}\right] \quad ; i \neq g$$

$$P_g = A_1\left[\frac{n}{(1+i)}\right] \quad ; i = g$$

EXERCISES



EASY

AN INVESTMENT PAYS $1,000 IN YEAR 1, INCREASING BY $200/YEAR FOR 5 YEARS. FIND THE PRESENT WORTH (P) AT I = 10%

MEDIUM

FACTORY MAINTENANCE IS $100K IN YEAR 1 AND INCREASES BY 8% ANNUALLY. FIND P OVER 10 YEARS AT I = 12%

CLINCHER

A COMPANY EXPECTS A PROFIT OF $1M THIS YEAR. THEY CAN INVEST IN SYSTEM A (LINEAR GROWTH: +$100K/YEAR) OR SYSTEM B (GEOMETRIC GROWTH: +6%/YEAR). OVER A 10-YEAR HORIZON AT 8% INTEREST,

WHICH SYSTEM YIELDS A HIGHER PRESENT WORTH?

AMORTIZATION

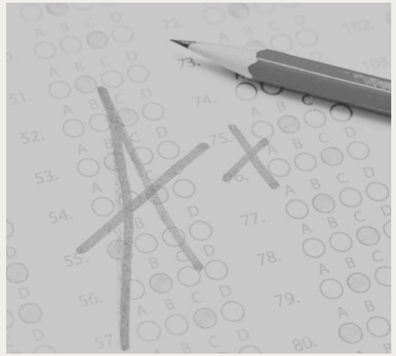
DEFINITION

- AMORTIZATION IS THE ACCOUNTING PROCESS USED TO SPREAD THE COST OF INTANGIBLE ASSETS OVER THE PERIODS EXPECTED TO BENEFIT FROM THEIR USE.
- THE CUSTOMARY METHOD FOR AMORTIZATION IS THE STRAIGHT-LINE METHOD.
- DETERMINING WHICH INTANGIBLE ASSETS MAY BE AMORTIZED AND THE CORRECT CAPITALIZED VALUE CAN SOMETIMES BE TRICKY.
- AMORTIZATION RULES DIFFER SIGNIFICANTLY FOR TAX VERSUS BOOK PURPOSES.
- APPLIED CORRECTLY, AMORTIZATION CAN RESULT IN SIGNIFICANT TAX SAVINGS.

FORMULA

$$Annual\ Amortization = \frac{Capitalized\ Cost}{Useful\ Life}$$

EXERCISES



EASY

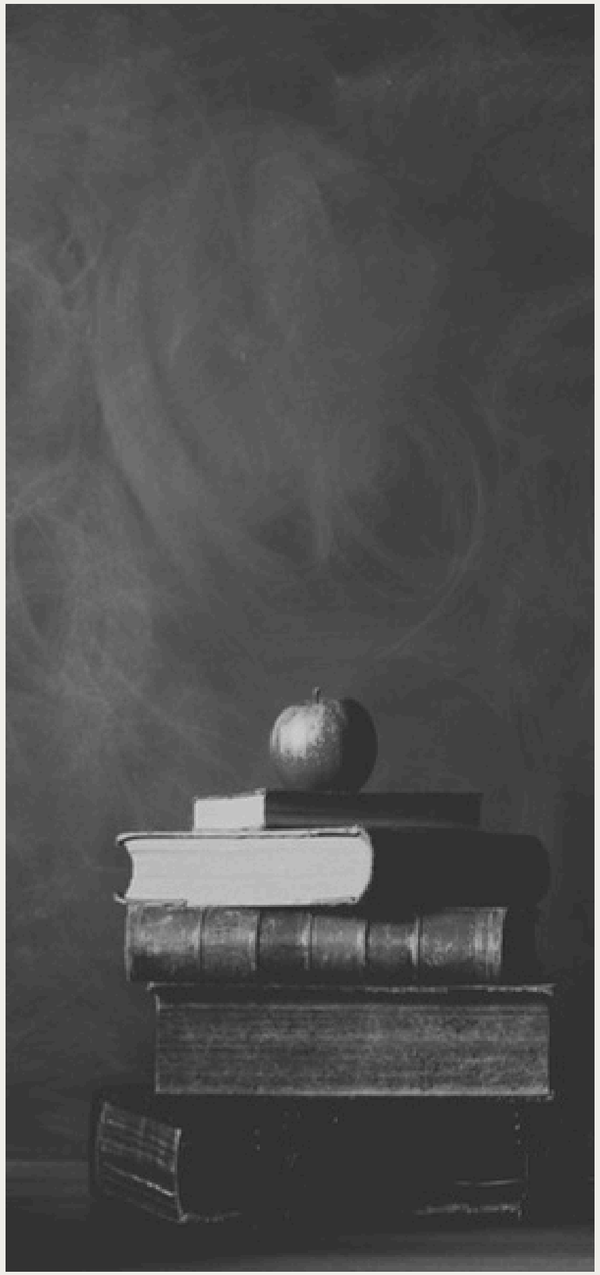
YOU BORROW $100,000 AT 12% ANNUAL INTEREST COMPOUNDED MONTHLY. WHAT IS THE MONTHLY PAYMENT FOR A 1-YEAR LOAN?

MEDIUM

FOR THE LOAN IN #1, WHAT IS THE REMAINING BALANCE AFTER THE 6TH PAYMENT? HOW MUCH INTEREST WAS PAID IN THE 7TH MONTH?

CLINCHER

A FIRM NEEDS $500K. BANK A OFFERS 10% FOR 5 YEARS. BANK B OFFERS 12% FOR 3 YEARS. IF THE FIRM'S GOAL IS TO MINIMIZE THE TOTAL INTEREST PAID OVER THE LIFE OF THE LOAN, WHICH BANK SHOULD THEY CHOOSE?



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